

SLR GROUP CHARTER & WORK BREAKDOWN REPORT

AI-Driven Model Engineering: A Systematic Literature Review of Intelligent Model Generation, Transformation, Validation, and Software Synthesis

1. Executive Summary & Research Motivation

Model-Driven Engineering (MDE) places models at the centre of software development, representing different system perspectives including context, interactions, structure, and behaviour. Conventional MDE traditionally transforms Computation-Independent Models (CIM) to Platform-Independent Models (PIM), then to Platform-Specific Models (PSM), and finally to executable software. Despite its benefits, MDE adoption remains limited by specialist skill requirements, tool complexity, rigid transformation rules, model-code inconsistencies, and the high effort needed to maintain models as requirements and environments evolve. Artificial intelligence—including large language models (LLMs), machine learning, deep learning, multimodal foundation models, knowledge graphs, neuro-symbolic methods, and intelligent agents—creates unprecedented opportunities to automate and augment model-engineering activities. The emerging research challenge is no longer limited to generating individual UML diagrams; it concerns how AI can support the complete, continuous, and trustworthy engineering of software models.

AI-Driven Model Engineering (AI-DME) is defined as the systematic application of artificial intelligence to generate, interpret, complete, transform, validate, optimise, synchronise, explain, and evolve software models throughout the software development lifecycle. AI-DME does not replace established modelling standards or human expertise. Instead, it introduces an intelligent layer that augments conventional model-engineering processes while remaining subject to deterministic validation and accountable human oversight. This project aims to systematically identify, classify, and critically evaluate AI applications across the model-engineering lifecycle to develop an evidence-based taxonomy, maturity model, evaluation hierarchy, and future research agenda for trustworthy AI-DME. To achieve this, a highly structured, fair, and rubric-aligned division of labor is established for our **7-member student group**.

CORE PROJECT PARAMETERS & SUBMISSION CONSTRAINTS

- **Academic Context:** Applications Development Theory 4 (APT470S) — NQF Level 7 Research Project (CPUT)
- **Submission Deadline:** 27 August 2026, Close of Business (CoB)
- **Synthesis Scope:** Search 6 primary databases (Scopus, Web of Science, IEEE Xplore, ACM DL, ScienceDirect, SpringerLink) from Jan 2015 to 2026
- **Evidence Dataset:** Initial retrieval: ~1,236 articles -> screened: 242 -> final inclusion: **86 papers** for qualitative synthesis
- **Formatting Rules:** Word (.docx) format, 20–30 pages (excluding references & appendices), Times New Roman, 11 pt, 1.5 line spacing, 1-inch margins, fully justified, APA 7th Edition referencing
- **Quality Mandate:** Methodologically rigorous, critically synthesised, 100% traceable, and zero plagiarised or AI-generated content

Proceed to the next page for the detailed 7-member group role assignments and rubric mapping.

2. 7-Member Group Structure & Role Assignments (Part 1)

A highly professional and fair division of labor is achieved by mapping roles directly to the 100-mark assessment rubric. Each member takes primary ownership of specific grading criteria while supporting administrative and operational team needs.

ROLE 1: Project Manager & Protocol Lead

Rubric Alignment: Criteria 1, 2, 3 (10 Marks Marks)

Core Focus: Overall scope management, timeline adherence, and formal SLR protocol definition.

Key Deliverables:

- Draft Section 1 (Research Significance & Background) [1].
- Define the research aim and align 15 Research Questions (RQs) [5, 7, 8].
- Develop and document the reproducible SLR protocol using SE guidelines & PRISMA [14].

Secondary Operational Role: Lead Paper Editor. Responsible for final compilation of the 20-30 page Word document, formatting (Times New Roman 11pt, 1.5 line spacing, 1-inch margins) [24].

ROLE 2: Search Specialist & Reference Librarian

Rubric Alignment: Criterion 4 (5 Marks Marks)

Core Focus: Executing database search queries, de-duplication, screening, and documentation.

Key Deliverables:

- Develop and adapt the detailed search string for the 6 primary databases [15].
- Apply systematic inclusion/exclusion criteria on the retrieved ~1,236 papers [16].
- Document the complete literature sifting process in a PRISMA flow diagram [25].

Secondary Operational Role: Reference Librarian. Manages the shared group bibliography in Mendeley/Zotero, ensuring flawless APA 7th Edition citations across the paper and tables [24].

ROLE 3: Quality Assessment & Bias Analyst

Rubric Alignment: Criterion 5 (10 Marks Marks)

Core Focus: Designing and executing the quality and risk-of-bias appraisal on all final studies.

Key Deliverables:

- Adapt the 17-item Quality Assessment Checklist covering AI & model transparency, dataset quality, validity, baselines, and scalability [17-19].
- Perform quality appraisal on all 86 included papers [25].
- Analyze threats to validity and risk-of-bias patterns across the literature [20].

Secondary Operational Role: Compliance & Academic Integrity Officer. Performs Turnitin screenings to ensure zero plagiarism, and drafts the mandatory AI-Usage Declaration [24, 41].

ROLE 4: Data Extraction Architect

Rubric Alignment: Criteria 6, 11 (20 Marks Marks)

Core Focus: Designing, pilot-testing, and managing the core data extraction database.

Key Deliverables:

- Build and pilot-test the Excel data extraction sheet with all 19 framework variables [20, 44].
- Oversee and cross-verify the data extraction executed by all team members [37].
- Ensure 100% data traceability back to primary papers and maintain extraction consistency [30, 37].

Secondary Operational Role: Visualizations & Data Analyst. Translates extracted database fields (such as publication year, AI techniques, modelling languages) into publication-quality charts [41].

2. 7-Member Group Structure & Role Assignments (Part 2)

Roles 5 through 7 take the lead in synthesis, original framework designs, and systems engineering, completing the core academic requirements for an Excellent (75-100%) grade.

ROLE 5: Lead Evidence Synthesizer

Rubric Alignment: Criterion 7 (15 Marks Marks)

Core Focus: Performing in-depth qualitative, thematic, and comparative synthesis of outcomes.

Key Deliverables:

- Synthesize publication trends, AI technologies, and model-engineering stages [21].
- Conduct thematic synthesis on core tasks (e.g., requirements-to-model, agentic workflows) [21].
- Develop comparative analyses: AI vs. human modellers, proprietary vs. open-source [21].

Secondary Operational Role: Slide Deck Committee. Coordinates with the group to outline and draft the final presentation slides in accordance with professional and academic standards [42].

ROLE 7: Systems Architect & Research Agenda Lead

Rubric Alignment: Criterion 9 (5 Marks Marks)

Core Focus: Proposing the reference architecture, analyzing gaps, and mapping future work.

Key Deliverables:

- Design a verifier-guided, standards-compliant, human-accountable reference architecture [22, 40].
- Construct the final literature Gap Matrix mapping covered areas vs. research gaps [24, 40].
- Draft a prioritized future research agenda for trustworthy and autonomous model engineering [22, 40].

Secondary Operational Role: Oral Presentation Lead. Authors the spoken script and coordinates the group rehearsal and division of speaking time for the final 10-mark oral assessment [42].

ROLE 6: Theoretical Framework Designer

Rubric Alignment: Criterion 8 (15 Marks Marks)

Core Focus: Designing and validating original scholarly frameworks based on the extracted evidence.

Key Deliverables:

- Develop a comprehensive, multidimensional taxonomy of AI-DME [12, 22].
- Define the 6 levels of the AI-DME Automation Continuum / Maturity Model [13, 22].
- Establish the 7-level Model Quality and Assurance Hierarchy [22].

Secondary Operational Role: Lead Graphic Designer. Responsible for designing all visual schemas, including the final Conceptual Framework diagram and taxonomy trees, ensuring professional style [24, 41].

SHARED CORE TEAM RESPONSIBILITIES

- **Academic Writing & Polish (Criterion 10 - 5 Marks):** All members must author their respective sections in clear, coherent, and concise academic English. Consistent terminology and flawless APA 7th referencing must be maintained throughout [24, 41].
- **Oral Professional Presentation (Criterion 12 - 10 Marks):** All 7 members must actively present during the final oral evaluation, demonstrating complete mastery of the study and responding confidently to Q&A [42].
- **Peer Contribution & Participation (Criterion 13 - 5 Marks):** Active attendance at weekly group sessions, on-time task submission, and collaborative data extraction [43]. Each member will extract data for approximately 12 primary studies to cover the 86 included papers [25].

3. SLR Pipeline & Cross-Role Handoff Checkpoints

To ensure seamless group coordination and prevent data bottlenecks, the research follows a strict, sequential pipeline. Each handoff point represents a formal contract of deliverables that must be signed off by the Project Manager [30].

Handoff Stage	Sender Role	Receiver Role	Contractual Deliverable / Handoff Asset
1. Literature Screening	Search Specialist (Role 2)	Quality Analyst (Role 3) & Data Architect (Role 4)	List of 86 peer-reviewed papers with full-text PDFs and basic citation metadata [25].
2. Extraction Framework	Data Architect (Role 4)	Entire Group (Roles 1–7)	Pilot-tested Excel extraction template with 19 defined coding variables and data definitions [20].
3. Coding & Appraisal	Entire Group (Roles 1–7)	Quality Analyst (Role 3) & Data Architect (Role 4)	Individual extraction rows (approx. 12 papers/member) with full text-evidence and citations [24, 25].
4. Quality-Scored Data	Quality Analyst (Role 3) & Data Architect (Role 4)	Lead Synthesizer (Role 5) & Framework Designer (Role 6)	Consolidated, verified Excel extraction database containing all 19 variables and 17-item quality appraisal scores [35, 37].
5. Findings & Gaps	Lead Synthesizer (Role 5)	Framework Designer (Role 6) & Systems Architect (Role 7)	Detailed narrative summary of descriptive, thematic, and comparative findings, and identified research gaps [21].
6. Scholarly Artefacts	Framework Designer (6) & Systems Architect (Role 7)	Project Manager (Role 1) & Entire Group	Completed original visual figures: taxonomy tree, 6 maturity levels, quality hierarchy, verifier-guided architecture, and Gap Matrix [22].
7. Final Compilation	Project Manager (Role 1)	Entire Group (Roles 1–7)	Compiled 20–30 page Microsoft Word document (.docx) and complete Mendeley bibliography for final group review [24].

4. The Data Extraction Framework (Excel Tracker Structure)

The Data Architect (Role 4) manages the core Excel database. In alignment with *SLR_Team_Tracker (5).pdf*, all members must extract the following 29 data fields for their assigned papers to ensure absolute empirical traceability [44, 45].

Category	Extracted Framework Variables (29 Fields from SLR Team Tracker)
Administrative Metadata	# (Record Number), Title, Extracted By, Year, Venue (Conference/Journal), Country/Institution, Study/Contribution Type (e.g., Tool, Method, Dataset, Evaluation) [20, 44].
Context & System Scale	Application Domain (e.g., Automotive, Health, FinTech), System Scale (e.g., Textbook, Industrial, Benchmark) [20, 44].
AI Technique & Architecture	AI Technique/Model/Version (e.g., GPT-4o, LLaMA-3, Graph Neural Network, Reinforcement Learning), Prompting/Training/Adaptation Method (e.g., Zero-shot, Few-shot, Fine-tuning, RAG), Single- or Multi-Agent [20, 44].
Model-Engineering Specs	Model-Engineering Stage/Task (e.g., Generation, Transformation, Validation, Repair, Synchronisation) [9, 10, 20], Modelling Language/Metamodel/Diagram Type (e.g., UML Class, Sequence, State, SysML, BPMN, Domain-Specific Language), Input Artefact(s), Output Artefact(s) [12, 20, 44].
Rigour, Baselines & Cost	Dataset/Benchmark, Baseline/Comparison Method (e.g., Human modellers, Rule-based MDE tools), Computational Cost/Scalability, Explainability/Uncertainty Treatment, Reproducibility Assets [20, 44].
Quality & Validation Specs	Validation Mechanism (e.g., OCL constraint check, Syntax parsing, Test execution, Human review) [12, 20], Human Involvement (e.g., None, In-the-loop, Post-review) [20, 44], Quality Metrics & Reported Performance, Standards Conformance/Traceability [20, 44].
Adoption & Maturity Outcomes	Reported Benefits, Reported Limitations, Industrial Evidence, Assigned Maturity Level (1–6) (Based on the AI-DME Automation Continuum / Maturity Model) [20, 45].

5. Rubric-Aligned Quality Checklists

This step-by-step checklist serves as our final quality assurance gate. To score in the **Excellent (75-100%)** grading band, the final compiled paper and data extraction spreadsheet must fulfill all points below [30].

RESEARCH SIGNIFICANCE & OBJECTIVES QUALITY CHECKS (Excellent Band Criteria)

- Provide a compelling, highly contextualized research significance for AI-DME backed by high-quality recent literature [31].
- Define MDE and AI-DME precisely, clarifying that AI does not replace standards but adds an intelligent layer [2].
- Align all 15 RQs and research objectives, ensuring scope and boundaries are precise and defensible [32].

METHODOLOGICAL RIGOUR & SEARCH STRATEGY QUALITY CHECKS (Excellent Band Criteria)

- Document a complete, highly reproducible SLR protocol based on recognized Software Engineering guidelines and the PRISMA framework [33].
- Define adapted search strings and key terms clearly, with explicit reasons for database selections [34].
- Apply inclusion/exclusion criteria consistently and document the entire selection using a PRISMA flowchart (1,236 down to 86 papers) [24, 25].
- Formulate and score a 17-item Quality Assessment Checklist (0 = No, 0.5 = Partial, 1 = Yes) on all 86 papers and weight evidence accordingly [17-19, 37].

DATA EXTRACTION & SYNTHESIS QUALITY CHECKS (Excellent Band Criteria)

- Utilize a pilot-tested Excel extraction template capturing AI tech, modelling tasks, inputs, outputs, validation, cost, and maturity [37, 44].
- Ensure 100% data traceability. Every single claim, data cell, and table entry must trace back to primary studies with citations [30, 37].
- Integrate descriptive, thematic, and comparative syntheses with strong critical reasoning and evidence-based analysis [38].
- Explicitly address under-researched areas and identify critical gaps (e.g., class diagrams overemphasis, weak cross-view consistency) [11].

SCHOLARLY CONTRIBUTION & FORMAT CONSTRAINTS QUALITY CHECKS (Excellent Band Criteria)

- Derive original, validated scholarly models: a detailed taxonomy, a 6-level Maturity Model, and a 7-level Quality Hierarchy [22, 39].
- Design a clear verifier-guided, standards-compliant, human-accountable reference architecture [22, 40].
- Draft a prioritized future research agenda and a literature Gap Matrix directly grounded in the review's evidence [22, 40].
- Maintain absolute document standards: Times New Roman 11pt, 1.5 line spacing, 1-inch margins, fully justified, 20-30 pages docx [24].
- Ensure consistent APA 7th Edition referencing (including within tables) and include the mandatory AI-Usage Declaration [24, 41].